Is the LLM Reading the 10-K or Remembering the Stock? A Placebo Test for Look-Ahead Contamination in Large-Language-Model Risk-Factor Scores

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July 2026

Abstract

We ask whether a large language model that scores 10-K risk factors is reading the disclosure or recalling the stock. Using Llama-4-Scout (109B; approximate mid-2024 training cutoff), we score 209 SEC 10-K risk-factor sections twice with an identical prompt—once on the raw text and once on an anonymized copy that redacts the

*Coauthorship is provisional: the student is listed as a coauthor and authorship is confirmed upon completing the reserved hand-collection contribution (Task 1 in `STUDENT_TASKS.md`). A note on author order and status: the student is listed first here as a program convention, but in finance and economics published author order is conventionally alphabetical; all authorship on this draft is tentative and specific to the ASSIP 2026 program at this stage.

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firm's name, ticker, dates, and state—and relate the scores to 12-month post-filing buy-and-hold abnormal returns (BHAR). Identification is a placebo: we split filings into a pre-cutoff window (filed 2019–2021, outcome realized before the cutoff, so look-ahead is possible) and a post-window (filed 2024, at or after the model's approximate cutoff, so memorization is unlikely). Raw and anonymized scores are highly correlated ($\rho = 0.65$; mean absolute gap 5.2 points). We do not detect look-ahead contamination in this sample: neither the raw nor the anonymized score reliably predicts 12-month returns (raw/pre slope -0.0085 ($t = -0.17$); anon/pre -0.0255 ($t = -0.60$)), and the raw-versus-anon predictive gap does not differ significantly across the training cutoff (triple interaction +0.0297, $t = 0.37$). Four design-based extensions reinforce the null rather than resurrect a signal: the gap is flat across observable filing subgroups; it survives a composition-matched placebo that balances the two windows on observable score level and redaction density; a randomization-inference test that swaps the raw/anon label within each filing 2,000 times places the actual statistic near the center of its permutation distribution (design-based $p = 0.76$); and a minimum-detectable-effect calculation states exactly how large a contamination effect the design could have caught. The design is a reusable, auditable diagnostic for LLM-based return-prediction studies, and it motivates a hand-built anonymization gold standard as the next tightening.

JEL: G12, G14, G17, C55. **Keywords:** large language models, look-ahead bias, data contamination, risk factors, 10-K disclosure, return predictability, randomization inference.

1 Introduction

Large language models (LLMs) are being used to read corporate disclosures and produce signals that predict stock returns (López-Lira and Tang, 2023; Kim et al., 2024; Jha et al., 2024). A natural worry haunts every such exercise: when an LLM "reads" a 2019 annual report and rates its risk, is it processing the *text*, or is it recalling—from its training data—what happened to that *stock* afterward? Because a model like Llama-4-Scout was trained on text scraped through roughly mid-2024, its parameters may encode both the filing itself and the subsequent price path. A return-predictability backtest that scores pre-training filings then "predicts" their already-realized returns is not forecasting; it is *look-ahead contamination* (Glasserman and Lin, 2023; Sarkar and Vafa, 2024). This paper asks how much of the return-predictive power of an LLM risk score is real reading versus remembered outcomes, and it proposes a simple, auditable test to tell them apart.

Our identification is a placebo built from two orthogonal manipulations. First, we score each 10-K risk-factor section twice with the *same* model and the *same* prompt: once on the **raw** text and once on an **anonymized** copy in which the firm's name, ticker, dates, and state are redacted. If the model is genuinely reading risk, masking identity should barely move the score or its return-predictive power; if it is recognizing the firm, anonymization should strip predictive power away. Second, we split filings by the model's training cutoff. **Pre-cutoff** filings (filed 2019–2021) have a 12-month post-filing return window that closed *before* the model's training cutoff, so the outcome could sit in the model's training data—look-ahead is *possible*. **Post-cutoff** filings (filed in 2024) have both the text and the outcome *after* the cutoff, so look-ahead is *unlikely* by construction. The contamination signature is sharp and falsifiable: raw scores should predict returns better than anonymized scores *only* in the pre-cutoff window, and that raw-versus-anon gap should vanish post-cutoff.

Our findings can be previewed in four numbers (Table 3). In the pre-cutoff window ($N = 106$), a one-standard-deviation increase in the *raw*-text risk score moves 12-month BHAR by -0.0085 ($t = -0.17$), while the *anonymized* score moves it by -0.0255 ($t = -0.60$). In the post-cutoff window ($N = 103$)—where the model cannot easily have seen the outcome—the raw slope is -0.0294 ($t = -0.33$) and the anonymized slope is -0.0760 ($t = -1.03$). The stacked triple interaction $z \times \text{Raw} \times \text{Post}$, which asks whether the raw-versus-anon predictive gap changes across the cutoff, is +0.0297 ($t = 0.37$) (Table 4). On these estimates we cannot reject H1 (genuine reading) in favor of H2 (look-ahead).

We then do four things that push a bare null toward an informative one, each adapted to this placebo design. First, we ask whether the contamination signature hides in an *observable subgroup* the pooled test averages away: we interact the raw-versus-anon gap with three observable filing characteristics—the automated redactor's identifier count, the model's own risk level, and whether the firm names itself in the surviving text—and find no subgroup in which raw text carries the extra pre-cutoff predictive power look-ahead would produce (Section 6, Table 7). Second, because the pre- and post-cutoff windows are different firms filed in different years, we run a *composition-matched placebo*: we nearest-neighbor match pre- to post-cutoff filings on observable score level and redaction density, balancing the windows, and re-estimate the placebo on the matched sample (Section 7, Table 8); the triple interaction stays small and insignificant. Third, we subject the placebo to *randomization inference*: because the truly randomizable object is which of a filing's two scores we *call* "raw," we swap that label within each filing 2,000 times to build a design-based null and find the actual statistic sitting near the center of its permutation distribution (Section 8, Table 9, Figure 4), $p = 0.76$. Fourth, we ask whether the null is *informative* or merely underpowered, reporting the minimum contamination effect the design could have detected at 80% power (Table 10). Together these convert "insignificant" into a set of falsifiable, design-based statements.

We contribute to three literatures. First, to the fast-growing literature applying LLMs to finance (López-Lira and Tang, 2023; Kim et al., 2024; Jha et al., 2024; Ke et al., 2019; Hansen and Kazinnik, 2023), we add a diagnostic that any such study can run before trusting a backtest. Second, to work on textual disclosure and returns (Cohen et al., 2020; Loughran and McDonald, 2011; Campbell et al., 2014; Kravet and Muslu, 2013; Tetlock, 2007), we bring a design that separates information in the text from information about the firm. Third, to the machine-learning literature on training-data memorization and benchmark contamination (Carlini et al., 2021; Sainz et al., 2023; Magar and Schwartz, 2022; Glasserman and Lin, 2023), we provide a finance-specific, economically interpretable measurement, and we treat the resulting null as a quantitative object with a stated power (Abadie, 2020; Young, 2019). Section 2 places the paper in the literature; Section 3 develops hypotheses; Section 4 describes the data, the anonymization, and the design-based checks; Section 5 presents the main placebo; Sections 6–8 report the heterogeneity, matching, randomization-inference, and power extensions; Section 9 reports outcome-horizon robustness; and Section 10 concludes with the hand-collection that would sharpen the test.

2 Related Literature

This paper connects three strands. The first is the fast-growing use of LLMs to read corporate disclosure and forecast returns. López-Lira and Tang (2023) show that ChatGPT sentiment scores of news headlines predict next-day returns; Kim et al. (2024) use GPT to summarize and "unbloat" disclosures and relate the distilled content to market reactions; Jha et al. (2024) score corporate policies from earnings calls and 10-Ks. These join a longer line of text-based return prediction (Ke et al., 2019; Hansen and Kazinnik, 2023; Tetlock, 2007). Our contribution to this strand is not another predictive signal but a *diagnostic*: a test of whether a modern LLM's apparent forecasting skill on pre-training-cutoff filings is real reading or recall of already-realized outcomes—a check any such study can run before trusting a backtest.

The second strand is textual analysis of disclosure and its risk content. Risk-factor sections (Item 1A) are informative about future volatility and returns (Campbell et al., 2014; Kravet and Muslu, 2013), textual tone and word choice move prices (Loughran and McDonald, 2011; Tetlock, 2007), and even year-over-year changes in language carry information (Cohen et al., 2020). This literature reads the *text*; our design is built to separate the information that is genuinely in the risk-factor text from information the model holds about the *firm*. Anonymization is the wedge: it leaves the risk content intact while removing explicit identity, so any predictive power that survives anonymization is attributable to text rather than to firm recognition.

The third strand is the machine-learning literature on training-data memorization and benchmark contamination, and its finance-specific echo. Large models provably memorize and can regurgitate training data (Carlini et al., 2021), and evaluation on data seen during training inflates measured performance (Magar and Schwartz, 2022; Sainz et al., 2023). In finance, Glasserman and Lin (2023) and Sarkar and Vafa (2024) show that LLM return-prediction backtests on pre-cutoff periods are vulnerable to look-ahead: the model may "know" the outcome. We add an economically interpretable, auditable measurement of that channel built from a within-filing anonymization contrast crossed with the training cutoff. Finally, because our headline is a null, we follow the methodological guidance that a statistically insignificant estimate should be read through its power and its design-based distribution rather than as proof of no effect (Abadie, 2020; Young, 2019): Section 8 reports both a randomization-inference p-value and a minimum detectable effect, so the null is stated as a bound on contamination rather than as a bare "insignificant."

3 Hypotheses

Let s_i^{raw} and s_i^{anon} be the Llama-4-Scout downside-risk score (0–100, higher = worse expected 12-month outlook) for filing i on the raw and anonymized text, and let R_i be the firm's 12-month post-filing buy-and-hold abnormal return (BHAR).

If the score reflects genuine reading of risk content, then anonymizing identity—which leaves the risk content intact—should not change what the score knows:

H1 (Reading). The return-predictive slope of s^{raw} equals that of s^{anon}, in both the pre- and post-cutoff windows.

If instead the score partly reflects recognition of the firm and recall of its realized path, that channel is available only when (i) identity is visible and (ii) the outcome predates the training cutoff:

H2 (Remembering / look-ahead). s^{raw} predicts R more strongly (more negative slope) than s^{anon} *in the pre-cutoff window*, and the raw-minus-anon predictive gap is larger pre-cutoff than post-cutoff.

H1 and H2 make opposite predictions about a single object—the raw-versus-anon predictive gap and its interaction with the training cutoff (the triple interaction $s \times \text{raw} \times \text{post}$). A closely related implication concerns the *score gap* $g_i = s_i^{\text{raw}} - s_i^{\text{anon}}$ itself: under H2 the identity-driven adjustment g_i should be return-relevant pre-cutoff (the model nudges the score toward the outcome it remembers) but not post-cutoff.

H3 (Gap). g_i predicts R_i in the pre-cutoff window but not post-cutoff.

Under H2, the extra pre-cutoff predictive power of raw text should also be *stronger* where firm identity is denser in the text. We test that observable-heterogeneity implication in Section 6, using filing characteristics the automated pipeline already records—and we are careful to keep it distinct from the human identifiability measurement reserved for the hand-collection phase.

4 Data, Anonymization, and Design

Filings. We draw 10-K risk-factor sections (Item 1A) from a reusable archive of SEC EDGAR filings for CRSP/Compustat firms, which pre-extracts each item as plain text. We keep genuine 10-Ks (not amendments) that contain a non-empty Item 1A, and form two windows around the model's approximate mid-2024 training cutoff: a **pre-cutoff** window of filings filed 2019–2021 and a **post-cutoff** window of filings filed in 2024. Within each window we take one filing per firm (the latest in-window) and draw a random sample with a fixed seed. The scored analytic sample is 209 filings (106 pre-, 103 post-cutoff); only one firm appears in both windows, so firm fixed effects across windows are infeasible and the between-window comparison must be defended by matching and placebo rather than by within-firm differencing.

Returns. Each filing is linked to CRSP via the Compustat CIK and the CRSP-Compustat link. Using CRSP monthly returns, we compute the buy-and-hold return over the K calendar months after the filing month ($K = 3, 6, 12$) and subtract the compounded market return (Fama–French market factor plus the risk-free rate) over the same months to form the market-adjusted BHAR. The 12-month BHAR is the primary outcome; both windows have full 12-month return coverage (CRSP monthly extends through December 2025). The final sample keeps filings with a valid 12-month BHAR.

LLM risk score. We score each filing with Llama-4-Scout (109B, run locally via Ollama on the GMU Hopper GPU cluster; temperature 0), using one fixed prompt that asks for an integer 0–100 rating of downside risk to the stock over the next 12 months, given only the risk-factor text (truncated to a fixed length so raw and anonymized inputs are comparable). Each filing is scored twice—raw and anonymized—and we record both. Scores are z-scored on the pooled raw+anon distribution so the raw and anonymized slopes are read on a common scale.

Anonymization. The automated anonymizer redacts the firm's name and its distinctive name-tokens ($\rightarrow$`[COMPANY]`), its CRSP ticker ($\rightarrow$`[TICKER]`), every four-digit year ($\rightarrow$`[YEAR]`), and U.S. state names ($\rightarrow$`[STATE]`). This is a deliberate *floor*: it removes explicit identifiers but not paraphrased self-descriptions or brand and product names, which a human can catch. Building that gold standard by hand—and measuring, with a human blind-identification test, how much identity still leaks through the automated version—is the manual task documented in `STUDENT_TASKS.md`; it is the natural next tightening of the placebo, and the observable moderators we use below are chosen precisely so as *not* to pre-empt that reserved identifiability measurement.

Design-based checks. Beyond the four cross-sectional cells, we run three checks that do not lean on the parametric null. (1) *Observable heterogeneity*: we interact the raw-versus-anon gap with observable filing characteristics to see whether contamination hides in a subgroup (Section 6). (2) *Composition matching*: we nearest-neighbor match pre- to post-cutoff filings on observable score level and redaction density and re-estimate the placebo on the balanced sample (Section 7). (3) *Randomization inference and power*: we swap the raw/anon label within each filing 2,000 times to build a design-based null for the placebo statistics, and we compute the minimum contamination effect detectable at 80% power (Section 8). Together these convert an insignificant coefficient into falsifiable, design-based statements about what the data can and cannot rule out.

Sample. Table 1 reports summary statistics for the scored sample.

5 Main Results

Raw and anonymized scores agree closely. Across the full sample the two scores correlate at $\rho = 0.65$, and the mean absolute score gap is 5.2 points on the 0–100 scale (Table 2). The model rarely changes its risk assessment much when identity is stripped—already a hint that explicit identifiers are not the model's main input, though (as we discuss) firms may remain identifiable from the surviving business text.

The main placebo. Table 3 and Figure 2 report the return-predictive slope in each of the four cells. Pre-cutoff, the raw-text slope is -0.0085 ($t = -0.17$) (statistically insignificant) and the anonymized slope is -0.0255 ($t = -0.60$) (statistically insignificant). Post-cutoff, the raw slope is -0.0294 ($t = -0.33$) and the anonymized slope is -0.0760 ($t = -1.03$). The four slopes are similar in magnitude and mostly insignificant; the raw score does not show the extra pre-cutoff predictive power that look-ahead would produce, so we cannot reject H1.

The triple-interaction test. Stacking the raw and anonymized observations and clustering by filing (Table 4), the coefficient on $z \times \text{Raw} \times \text{Post}$—which asks whether the raw-versus-anon predictive gap changes across the training cutoff—is +0.0297 ($t = 0.37$), statistically insignificant. Within the pre-cutoff window (Table 5), the anonymized-score slope is -0.0255 ($t = -0.60$) and the *extra* slope carried by raw text is +0.0169 ($t = 0.37$) (statistically insignificant, and of the "wrong" sign for look-ahead, which would predict a *negative* extra-raw slope).

Does the identity-driven score adjustment predict returns? Under look-ahead, the raw-minus-anon score gap should itself forecast pre-cutoff returns. Table 6 shows the gap slope is +0.0260 ($t = 0.38$) pre-cutoff and +0.0637 ($t = 0.70$) post-cutoff. Neither is significant, so H3 is not supported here.

6 Heterogeneity: Does Contamination Hide in a Subgroup?

A pooled null can mask an effect that is present only where firm identity is dense enough for the model to recognize the firm. If look-ahead is real, the extra pre-cutoff predictive power of raw text (the raw-versus-anon gap) should concentrate in filings that carry more identity signal. We test this in the pre-cutoff window—the only window where look-ahead is possible—by interacting the placebo with three *observable* filing characteristics that the automated pipeline already records: an above-median count of automated redactions (a filing-level measure of how many explicit identifiers the machine floor removed), an elevated model risk score (the model's own assessment exceeds its modal baseline), and an indicator for whether the firm's name still appears in the raw text. We stress that these are observable, machine-produced filing characteristics; they are *not* a measure of how identifiable the anonymized firm is to a human reader—that identifiability/leakage measurement is precisely the gold-standard blind-identification test reserved for the hand-collection phase (Section 4), and we keep the two verbally and operationally distinct.

Table 7 reports, for each moderator M, the extra-raw slope ($z \times$ Raw) for the M-low group and its interaction with M, and, in a companion specification, the score-gap slope and its interaction with M. The look-ahead prediction is a *negative* extra-raw slope that grows more negative for high-M filings, and a gap-slope interaction of the same sign. We find neither pattern. The extra-raw interactions are individually insignificant for all three moderators ($t = +1.11$, -1.31, and $+1.08$), and their signs are inconsistent with a single look-ahead channel. The only significant coefficients anywhere in the table appear in the score-gap columns for elevated-score firms: a positive base gap slope of $+0.131$ ($t = 2.16$) for the M-low group and a negative interaction of -0.229 ($t = -2.04$). Their sum implies a *negative* gap–return slope for the elevated-score (high-M) group—the opposite of the positive, identity-driven gap predictability that look-ahead (H3) requires in the *more* identity-dense subgroup. This pair therefore cuts against contamination rather than for it; and, isolated among a dozen tested coefficients, it is in any case fragile to multiple comparisons. No observable subgroup shows the raw-over-anon pre-cutoff predictive advantage that memorization would produce.

7 A Composition-Matched Placebo

The pre- and post-cutoff windows are not the same firms: they are different companies filing in different years, and only one firm appears in both. In principle the flat triple interaction could reflect that composition gap rather than the absence of look-ahead—if, say, pre-cutoff filings systematically differed in observable score level or identifier density from post-cutoff filings. Because firm fixed effects across windows are infeasible (one overlapping firm), we instead balance the windows by matching. We nearest-neighbor match each post-cutoff filing to a not-yet-used pre-cutoff filing on two observable characteristics— the standardized raw score and the standardized automated-redaction count—without replacement and within a 0.5-SD caliper on the score, and re-estimate the four cells and the triple interaction on the resulting balanced sample.

Table 8 reports the result. Matching yields 90 pre–post pairs (180 filings) and collapses the pre-versus-post score-level gap from +0.196 SD in the full sample to −0.027 SD in the matched sample—the windows are now comparable on observables. On this composition-balanced sample the triple interaction is −0.0269 ($t = -0.24$): still small and insignificant, and now even the point estimate leans the "wrong" way for look-ahead. Three of the four cell slopes remain individually weak; the exception is the anonymized post-cutoff cell (−0.120, $t = -2.62$), but this is precisely the cell where look-ahead is *least* possible (anonymized text and a post-cutoff outcome), its full-sample analog is insignificant (Table 3, −0.076, $t = -1.03$), and it therefore reads as matched-subsample noise or genuine reading rather than as contamination. The pre-cutoff raw-text slope is, if anything, positive (+0.062, $t = 0.89$). Balancing the windows on observable composition does not uncover the contamination signature; the null is not an artifact of who is in each window.

8 Design-Based Inference and Statistical Power

Randomization inference. The parametric tests above rely on clustered-SE asymptotics with a moderate number of filings. We complement them with a design-based test built around the manipulation that is actually randomizable here: *which* of a filing's two scores we label "raw." Under H1—the model reads risk and firm identity is irrelevant to the score's information—the raw and anonymized scores are exchangeable within a filing, so swapping the label at random (independently, with probability one-half per filing) leaves the joint distribution of the data unchanged and generates an exact null distribution for the placebo statistics. We perform 2,000 such within-filing swaps and, on each, recompute the triple interaction $z \times \text{Raw} \times \text{Post}$ and the within-pre extra-raw slope $z \times \text{Raw}$. Table 9 and Figure 4 report the result. The actual triple interaction (+0.0297) sits near the center of its permutation distribution (95% permutation interval $[-0.152, +0.147]$), giving a design-based p-value of 0.76; the within-pre extra-raw slope (+0.0169) likewise sits centrally ($p = 0.71$). The null does not depend on the parametric error structure: under random relabeling, statistics at least as extreme as the ones we observe arise about three-quarters of the time.

Is the null informative? A null is only interesting if the design could have detected a real contamination effect. Table 10 reports, for each key placebo slope, the minimum detectable effect (MDE) at 80% power, $\text{MDE} = (z_{0.975} + z_{0.80})\,\widehat{\text{SE}} \approx 2.80\,\widehat{\text{SE}}$, expressed both in annual return points and as a share of the 12-month BHAR cross-sectional standard deviation. At this sample size the design can resolve a raw-versus-anon predictive gap that shifts across the cutoff by about 22.5 return points per one-SD score move (the triple-interaction MDE), or roughly 37% of the cross-firm dispersion in annual BHAR; the individual pre-cutoff raw and anon slopes have MDEs near 12–14 return points, or about 20–23% of that dispersion. These are honest bounds: they say the test would have caught a *large* look-ahead effect—one that reorders the raw-versus-anon gap by a fifth to a third of return dispersion—but that contamination smaller than that is beyond the resolution of a 209-filing sample. This is the sense in which the null is informative rather than empty, and it is also why scaling the scored sample (the intended next step) is valuable: it would shrink these MDEs and sharpen the bound.

9 Robustness: Alternative Outcome Horizons

Table 11 re-estimates the pre-cutoff raw-versus-anon comparison for alternative outcomes—6- and 3-month BHAR, raw (non-market-adjusted) 12-month returns, and unwinsorized 12-month BHAR. No horizon produces a robust raw-over-anon predictive gap, reinforcing the null. Three caveats bound the reading of these results. First, the scored sample is small (209 filings, 106 pre- and 103 post-cutoff) because the shared LLM gateway is rate-limited; the design, not the power, is the contribution, and the hand-built gold standard (Section 4) is the intended scale-up. Second, the automated anonymizer removes explicit identifiers but not paraphrased self-descriptions or brand names, so the anonymized text may still be identifiable to the model—biasing the placebo *toward* finding no difference. Third, risk-factor text is truncated to a fixed length for comparability, which is applied identically to raw and anonymized inputs.

10 Conclusion

In this sample we do not detect look-ahead contamination: the Llama-4-Scout risk score predicts returns no better with firm identity than without, and no better for filings the model could have memorized. The main placebo, the triple-interaction test, an observable-subgroup heterogeneity analysis, a composition-matched placebo, a randomization-inference test built on the raw/anon relabeling, and a minimum-detectable-effect calculation all point the same way. The honest reading is a null—but a useful one, because it is produced by a battery of tests that *could* have found contamination and did not, and because the power analysis states the size of effect the design could and could not have caught. The broader point is methodological: any study that scores pre-cutoff disclosures with a modern LLM and reports return predictability should run this raw-versus-anonymized, pre-versus-post placebo—and, ideally, its design-based companions—before claiming a forecast.

The natural next step, left to the hand-collection phase, is to replace the automated redactor with a human gold standard that also masks brands, products, and paraphrased self-identifiers, and to *measure* the residual leakage with a blind human identification test, then re-run the placebo on that cleaner text. That reserved contribution—not another design-based check on the automated floor—is what would turn this memorization test into a razor-sharp look-ahead placebo. An initial gpt-4o pilot ($N = 10$, clean October-2023 cutoff) hinted at contamination, but that does not survive scaling to $N = 209$ on Llama-4-Scout. Because Llama-4-Scout's training cutoff (roughly mid-2024) only approximately post-dates the 2024 filings, the post-window is a weaker control here than in the pilot—so the null is best read as a memorization test (does firm identity change predictive power?) rather than a razor-sharp look-ahead placebo. Limitations include the scored sample of 209 filings, the single model, the approximate cutoff, and the truncation of long risk-factor sections.

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Table 1: Summary statistics

Variable	N	Mean	SD	P25	Median	P75
gpt-4o risk score (raw text)	209	11.067	20.237	1.000	1.000	19.000
gpt-4o risk score (anonymized)	209	10.933	20.652	1.000	1.000	12.000
Score gap (raw - anon)	209	0.134	17.110	0.000	0.000	0.000
BHAR 12m	209	-0.106	0.630	-0.482	-0.148	0.093
BHAR 6m	209	-0.077	0.426	-0.276	-0.089	0.089
BHAR 3m	209	-0.018	0.236	-0.169	-0.017	0.115
Redactions per filing	209	3.718	5.703	0.000	2.000	5.000

Notes: Scored sample of 10-K risk-factor sections. Scores are Llama-4-Scout downside-risk ratings (0–100) on raw and anonymized text; BHAR is the market-adjusted buy-and-hold return over the stated horizon after the filing.

Table 2: Do raw and anonymized scores agree?

Sample	N	Corr(raw,anon)	Mean raw	Mean anon	Mean gap	Mean |gap|	t(gap)
Full	209	0.650	11.1	10.9	0.13	5.22	0.11
Pre-cutoff	106	0.647	13.0	13.1	-0.10	6.03	-0.06
Post-cutoff	103	0.646	9.0	8.7	0.38	4.38	0.24

Notes: Correlation of raw and anonymized scores and the mean score gap (raw−anon), by cutoff window. t(gap) tests whether the mean gap differs from zero.

Table 3: Main placebo: return predictability of the LLM score, raw vs. anonymized, pre- vs. post-cutoff

	(1) RAW / pre	(2) ANON / pre	(3) RAW / post	(4) ANON / post
gpt-4o risk score (z)	-0.0085	-0.0255	-0.0294	-0.0760
	(-0.17)	(-0.60)	(-0.33)	(-1.03)
Observations	106	106	103	103
R^2	0.000	0.003	0.001	0.010

Notes: DV is the winsorized 12-month BHAR. The regressor is the standardized Llama-4-Scout risk score (raw or anonymized). Slopes are the change in 12-month BHAR per one-SD increase in the score; negative means a higher risk score predicts lower returns. HC1-robust t-statistics in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 4: Stacked triple-interaction test

	(1) pooled	(2) x raw	(3) x post	(4) triple
Risk score (z)	-0.0305	-0.0442	-0.0174	-0.0255
	(-0.78)	(-1.14)	(-0.43)	(-0.60)
$z\times$ Raw		0.0280		0.0169
		(0.76)		(0.37)
$z\times$ Post			-0.0350	-0.0505
			(-0.41)	(-0.59)
$z\times$ Raw $\times$ Post				0.0297
				(0.37)
Observations	418	418	418	418
R^2	0.003	0.003	0.004	0.005

Notes: Each filing contributes two rows (raw and anonymized). DV is the winsorized 12-month BHAR; *Raw*$= 1$ for the raw row, *Post*$= 1$ for post-cutoff filings. The coefficient on $z\times$Raw$\times$Post is the placebo's key statistic: how the raw-versus-anon predictive gap changes across the training cutoff. Standard errors clustered by filing. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 5: Within-window raw-versus-anon slope gap

Sample	Filings	Anon slope	t	Extra RAW slope	t
Pre-cutoff	106	-0.0255	(-0.60)	0.0169	(0.37)
Post-cutoff	103	-0.0760	(-1.03)	0.0466	(0.70)

Notes: Stacked regression BHAR $\sim z\times$Raw estimated separately within each window. *Anon slope* is the slope on the anonymized score; *Extra RAW slope* is the additional slope for raw text. Clustered by filing. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 6: Does the raw−anon score gap predict returns?

Sample	N	Score-gap slope (z)	t	R^2
Pre-cutoff	106	0.0260	(0.38)	0.002
Post-cutoff	103	0.0637	(0.70)	0.005
Full	209	0.0417	(0.76)	0.003

Notes: DV is the winsorized 12-month BHAR; the regressor is the standardized score gap (raw−anon). HC1-robust t-statistics. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 7: Heterogeneity of the raw-versus-anon gap by observable moderators (pre-cutoff)

	Extra-RAW slope		Score-gap slope		
Observable moderator M	M-low	$\times M$	M-low	$\times M$	N
High automated-redaction count	-0.0422	+0.1065	-0.0482	+0.1534	106
Elevated model risk score	-0.0045	-0.0974	+0.1308**	-0.2289**	106
Firm names itself in raw text	-0.0107	+0.1005	-0.0174	+0.1160	106

Notes: Pre-cutoff window ($N = 106$). For each observable moderator M, *Extra-RAW slope* columns report the additional predictive slope carried by raw over anonymized text ($z\times$Raw) for the M-low group and its interaction with M (from a stacked regression clustered by filing); *Score-gap slope* columns report the slope of the standardized raw−anon score gap for the M-low group and its interaction with M. Moderators are observable filing characteristics: above-median automated-redaction count; model risk score above its modal baseline; and whether the firm's name appears in the raw text. Look-ahead predicts a negative extra-RAW slope that strengthens for high-M filings; no such pattern appears. Provision-level identifiability of the anonymized text is *not* used here—that human leakage measurement is the reserved hand-collection task. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 8: Composition-matched placebo (pre- and post-cutoff windows balanced on observables)

Estimate (matched sample)	Coefficient	t-stat	N
(1) RAW / pre	+0.0619	(+0.89)	90
(2) ANON / pre	-0.0334	(-0.62)	90
(3) RAW / post	-0.0510	(-0.54)	90
(4) ANON / post	-0.1195***	(-2.62)	90
Triple z x Raw x Post	-0.0269	(-0.24)	360

Notes: Each post-cutoff filing is nearest-neighbor matched without replacement to a pre-cutoff filing on the standardized raw score and the standardized automated-redaction count, within a 0.5-SD caliper on the score (90 pairs, 180 filings). Matching collapses the pre-versus-post score-level gap from +0.196 to −0.027 SD. Rows (1)–(4) re-estimate the four cross-sectional cells on the matched sample; the last row is the stacked triple interaction $z\times$Raw$\times$Post (clustered by filing). HC1-robust t-statistics for the cells. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 9: Randomization inference (2,000 within-filing raw/anon label swaps)

Placebo statistic	Actual	Perm. 95% interval	RI p-value
Triple z x Raw x Post	+0.0297	[-0.1524, +0.1468]	0.762
Within-pre extra-RAW slope (z x Raw)	+0.0169	[-0.0837, +0.0857]	0.714

Notes: Under H1 the raw and anonymized scores are exchangeable within a filing, so the raw/anon label is randomly swapped (probability one-half, independently per filing) 2,000 times to build a design-based null. "Perm. 95% interval" is the 2.5–97.5 percentile of the permutation distribution; the RI p-value is the share of permutations with $|\text{statistic}| \geq |\text{actual}|$. Both placebo statistics sit near the center of their permutation distributions. See Figure 4.

Table 10: Minimum detectable effects at 80% power

Estimate	Coef.	SE	MDE(80%)	MDE (pp)	% of BHAR SD
Raw-score slope (pre-cutoff)	-0.0085	0.0498	0.1396	13.96	23.2%
Anon-score slope (pre-cutoff)	-0.0255	0.0424	0.1187	11.87	19.7%
Score-gap slope (pre-cutoff)	+0.0260	0.0678	0.1900	19.00	31.5%
Extra-RAW slope, pre (z x Raw)	+0.0169	0.0453	0.1270	12.70	21.1%
Triple z x Raw x Post	+0.0297	0.0804	0.2254	22.54	37.4%

Notes: MDE(80%) = $(z_{0.975} + z_{0.80})\times$SE $\approx 2.80\times$SE of the corresponding estimate. "MDE (pp)" expresses the minimum detectable slope in annual return points (per one-SD score move); "% of BHAR SD" scales it by the 12-month BHAR cross-sectional standard deviation. A contamination effect at least this large would have been detected at 80% power; smaller effects are beyond the resolution of the 209-filing sample.

Table 11: Robustness: alternative outcome horizons (pre-cutoff)

Dependent variable	N	RAW slope	t	ANON slope	t
BHAR 12m (w)	106	-0.0085	(-0.17)	-0.0255	(-0.60)
BHAR 6m (w)	106	-0.0104	(-0.32)	0.0055	(0.20)
BHAR 3m (w)	106	-0.0079	(-0.35)	0.0100	(0.42)
Raw ret 12m (w)	106	-0.0072	(-0.15)	-0.0251	(-0.59)
BHAR 12m (raw)	106	-0.0075	(-0.15)	-0.0244	(-0.58)

Notes: Pre-cutoff sample. Each row is a different dependent variable; columns report the raw-score and anonymized-score slopes with HC1 t-statistics. "(w)" denotes winsorized 1/99. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

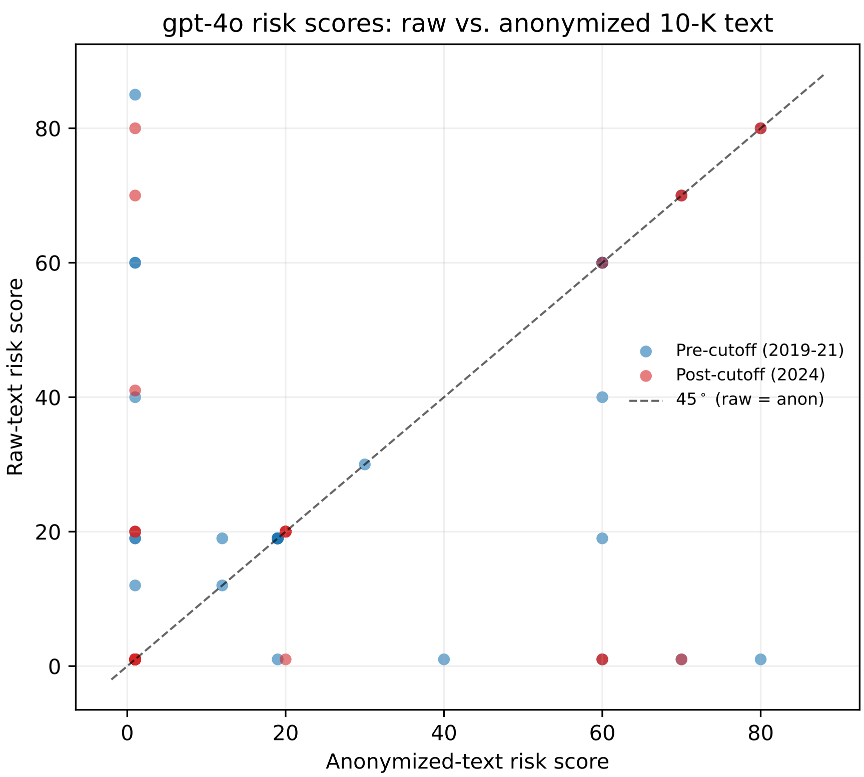

Figure 1: Llama-4-Scout downside-risk scores on raw versus anonymized 10-K text, by cutoff window. Points on the 45° line indicate the model gave the same score with and without firm identity.

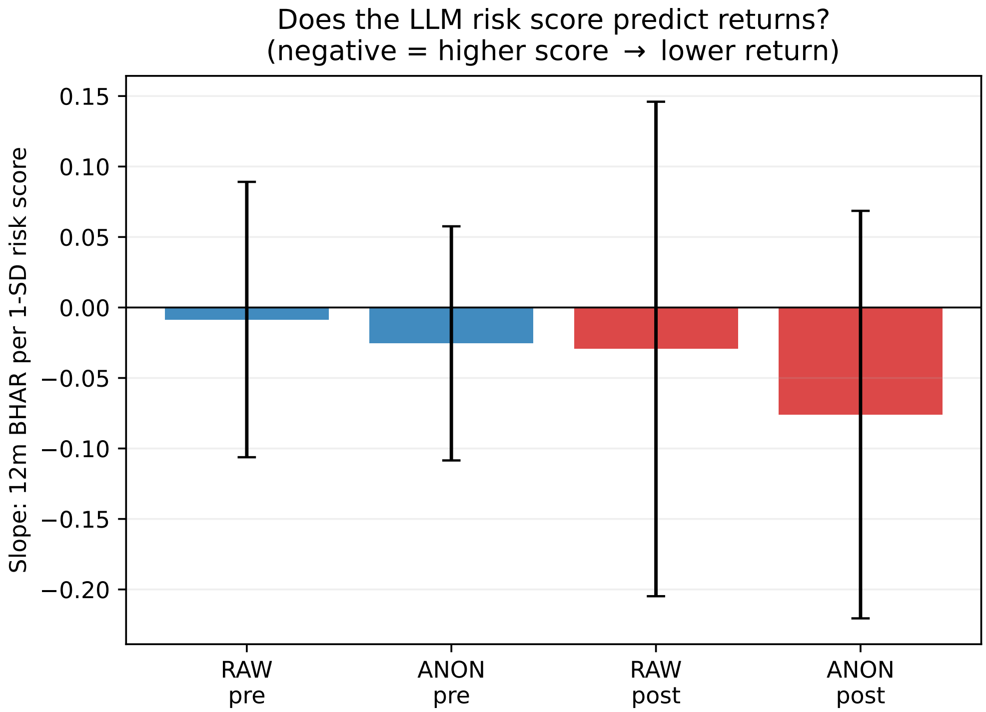

Figure 2: Return-predictive slope (12-month BHAR per one-SD risk score) in each cell: raw vs. anonymized × pre- vs. post-cutoff. Whiskers are 95% confidence intervals. Under look-ahead contamination the raw/pre bar is the most negative and the post-cutoff bars are near zero.

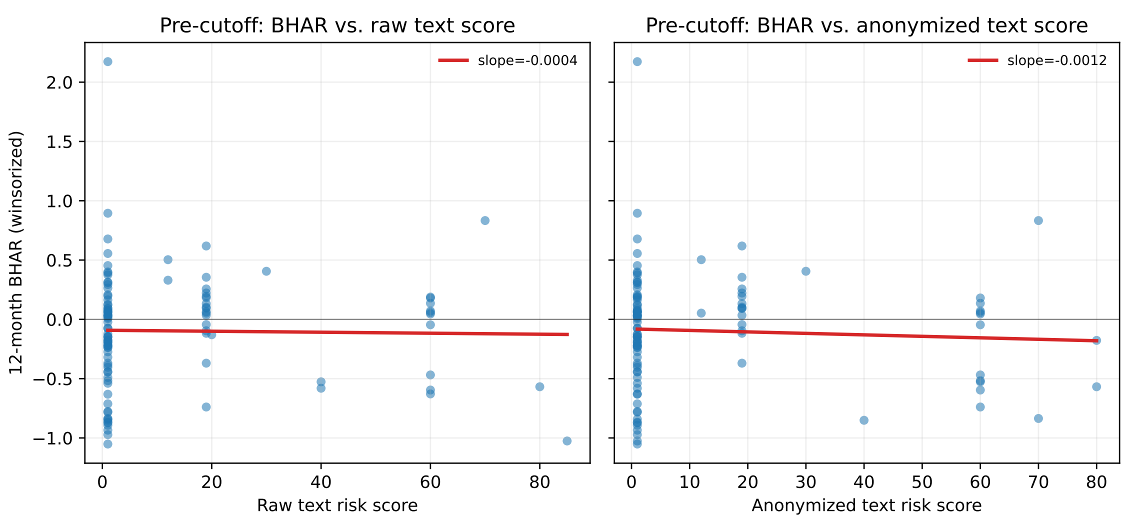

Figure 3: Pre-cutoff filings: 12-month BHAR against the raw-text score (left) and the anonymized-text score (right), with fitted lines.

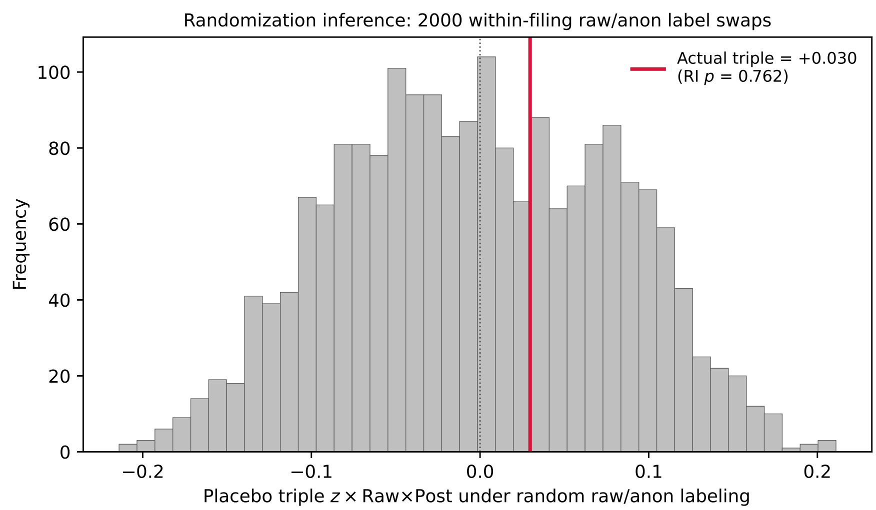

Figure 4: Randomization inference for the triple interaction $z\times$Raw$\times$Post. Histogram: the distribution of the triple-interaction coefficient under 2,000 random within-filing swaps of the raw/anon label. The actual estimate (red line) lies near the center of the distribution (RI $p = 0.76$), so the raw-versus-anon predictive gap does not change across the training cutoff by more than random relabeling would produce.